The documentation and process conversion measures necessary to comply with this revision shall be completed by 27 December 1999.

INCH POUND
MIL-PRF-19500/317G
27 September 1999
SUPERSEDING
MIL-S-19500/317F
16 September 1993

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING,
TYPES 2N2369A, 2N3227, 2N4449, 2N2369AU, 2N3227U, 2N4449U,
2N2369AUA, 2N3227UA, 2N4449UA, 2N2369AUB, 2N3227UB, and 2N4449UB
JAN JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, high speed switching transistors. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (TO-18) for 2N2369A and 2N3227, figure 2 (similar to TO-46) for 2N4449, figure 3 (similar to C-4) for UA version, figure 4 (similar to C-3) for UB version, and figure 5 (similar to C-6) for U version

1.3 Maximum ratings.

Types	P_T T_A = +25°C	P_T T_C = +25°C	V_{CBO}	V_{EBO}	V_{CEO}	V_{CES}	T_{OP} and T_{STG}
	W	W	V dc	V dc	V dc	V dc	°C
2N2369A	0.36 1/	1.2 2/	40	4.5	15	40	-65
2N3227	0.36 1/	1.2 2/	40	6.0	20	40	to
2N4449	0.30 3/	1.5 4/	40	4.5	15	40	+200

1/ Derate linearly 2.06 mW/°C for T_A > +25°C.
2/ Derate linearly 6.85 mW/°C for T_C > +25°C.
3/ Derate linearly 1.71 mW/°C for T_A > +25°C.
4/ Derate linearly 8.56 mW/°C for T_A > +25°C.

1.4 Primary electrical characteristics.

Type	h_{FE2} 1/ V_{CE} = 0.4 V dc I_C = 30 mA dc		h_{FE4} 1/ V_{CE} = 1.0 V dc I_C = 100 mA dc		$R_{\theta JC}$ °C/mW	
	Min	Max	Min	Max	Min	Max
2N2369A	30	120	20	120	- - -	146
2N3227	30	120	20	120	- - -	117
2N4449	40	250	30	150	- - -	146

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Primary electrical characteristics continued.

Limits	$\|h_{FE}\|$ V_{CE} = 10 V dc I_C = 10 mA dc f = 100 MHz		$V_{CE(sat)1}$ I_C = 10 mA dc I_B = 1 mA dc	t_{on} I_C = 10 mA dc I_{B1} = 3 mA dc I_{B2} = -1.5 mA dc	t_{off} I_C = 10 mA dc I_{B1} = 3 mA dc I_{B2} = -1.5 mA dc	t_s
	Min	Max	V dc	ns	ns	ns
2N2369A	5.0	10	0.20	12	18	13
2N4449	5.0	10	0.20	12	18	13
2N3227	5.0	10	0.20	12	25	13

1/ Pulsed (see 4.5.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Performance Specification Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service, 700 Robbins Avenue, Building 4D (DPM-DODSSP), Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.5).

3.2 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and herein.

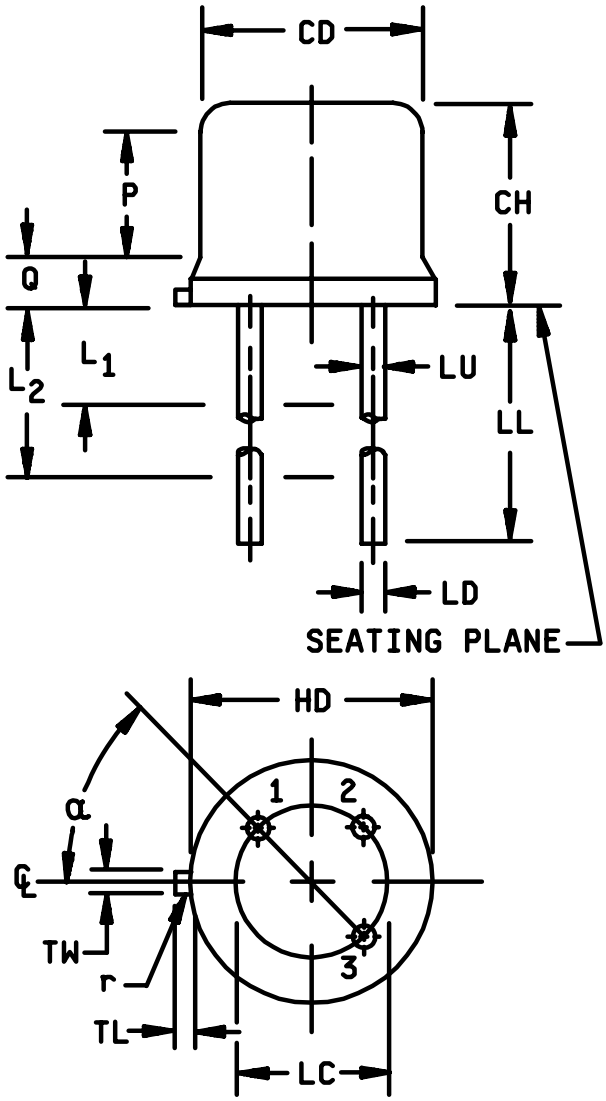
3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I herein. When a pacticular device is specified, the limit applies to all package types (e.g., 2N2369A, 2N2369AU, 2N2369AUA, and 2N2369AUB)

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Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.170	.210	4.32	5.33	
HD	.209	.230	5.31	5.84	
LC	.100 TP		2.54 TP		6
LD	.016	.021	0.41	0.53	7,8
LL	.500	.750	12.70	19.05	7,8,13
LU	.016	.019	0.41	0.48	7,8
L1	---	.050	---	1.27	7,8
L2	.250	---	6.35	---	7,8
Q	---	.030	---	0.76	5
TL	.028	.048	0.71	1.22	3,4
TW	.036	.046	0.91	1.17	3
r	---	.010	---	0.25	10
α	45° TP		45° TP		6

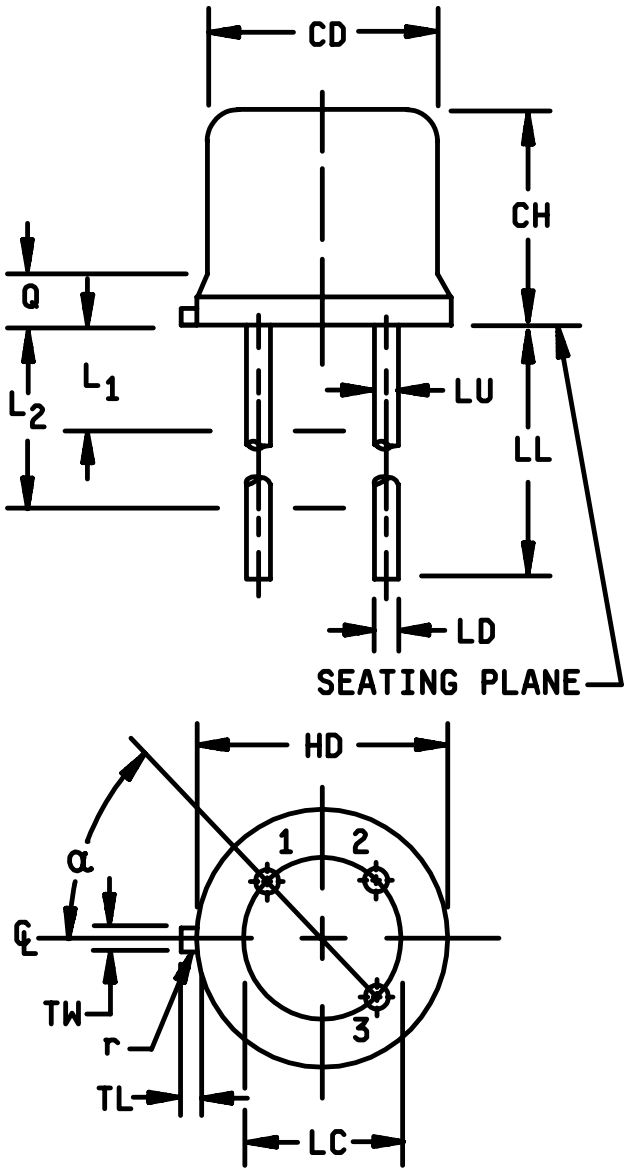


NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TH shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedure shown in figure 2.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.
12. Lead 1 = emitter, lead 2 = base, lead 3 = collector.

FIGURE 1. Physical dimensions (TO-18) (2N2369A and 2N3227)

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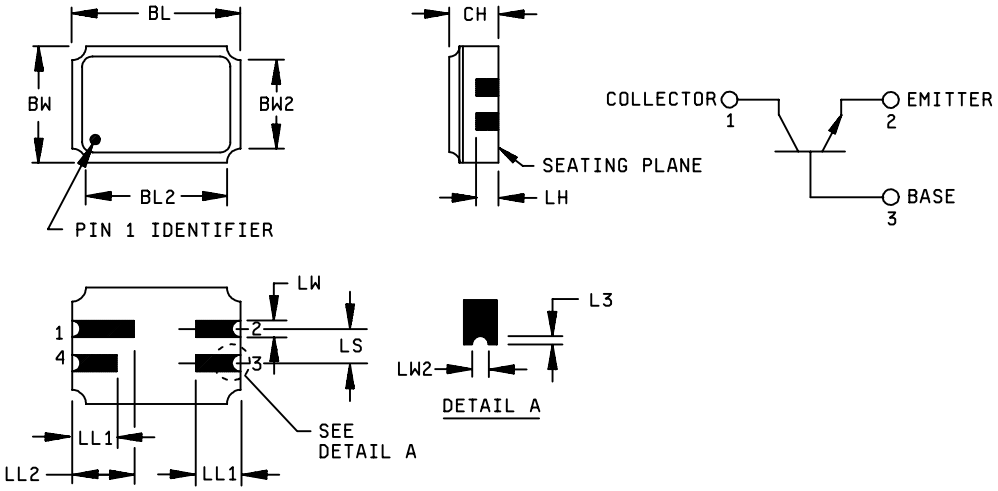
Symbol	Inches		Millimeters		Note
	Min	Max	Min	Max	
CD	0.178	0.195	4.52	4.95	
CH	0.065	0.085	1.65	2.16	
HD	0.209	0.230	5.31	5.84	
LC	0.100 TP		2.54 TP		6
LD	0.016	0.021	0.41	0.53	7
LL	0.500	0.750	12.70	19.05	7
LU	0.016	0.019	0.41	0.48	7
L_1		0.050		1.27	7
L_2	0.250		6.35		7
TL	0.028	0.048	0.71	1.22	3
TW	0.036	0.046	0.91	1.17	10
Q		.040		1.02	4
r		0.007		0.18	11
α	45° TP				6



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Symbol TL is measured from HD maximum.
4. Details of outline in this zone are optional.
5. Symbol CD shall not vary more than 0.010 inch (0.25 mm) in zone P. This zone is controlled for automatic handling.
6. Leads at gauge plane 0.054 inch (1.37 mm) +0.001 inch (0.03 mm) -0.000 inch (0.00 mm) below seating plane shall be within 0.007 inch (0.18 mm) radius of TP relative to tab. Device may be measured by direct methods or by gauge.
7. Symbol LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Lead diameter shall not exceed 0.042 inch (1.07 mm) within L_1 and beyond LL minimum.
8. Lead #1 = emitter, lead #2 = base, lead #3 = collector.
9. Lead number three is electrically connected to case.
10. Beyond r maximum, TW shall be held for a minimum length of 0.011 inch (0.28 mm).
11. Symbol r applied to both inside corners of tab.

FIGURE 2. Physical dimensions (TO-46) 2N2309A and 2N3227.

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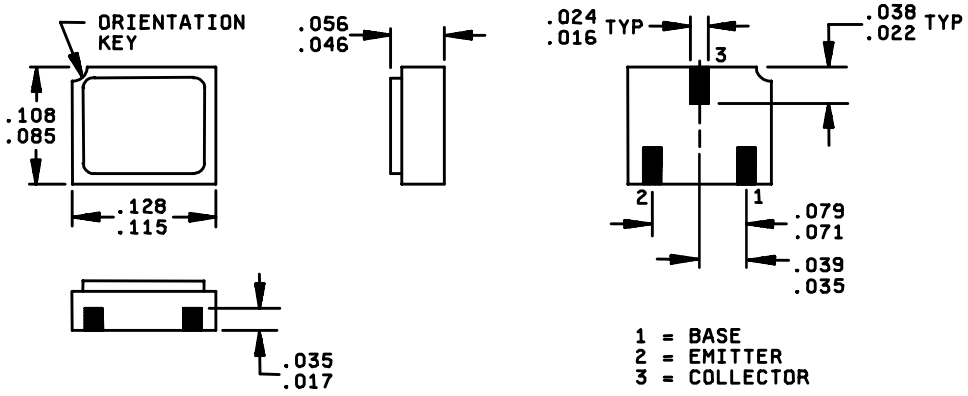
Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.215	.225	5.46	5.71
BL_2		.225		5.71
BW	.145	.155	3.68	3.93
BW_2		.155		3.93
CH	.061	.075	1.55	1.90
L_3	.003	.007	0.08	0.18
LH	.029	.042	0.74	1.04
LL_1	.032	.048	0.81	1.22
LL_2	.072	.088	1.83	2.23
LS	.045	.055	1.14	1.39
LW	.022	.028	0.56	0.71
LW_2	.006	.022	0.15	0.56

Transistor	Collector	Emiiter	Base	N/C
Pin no.	1	2	3	4

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The coplanarity deviation of all terminal contact points, as defined by the device seating plane, shall not exceed .006 inch (0.15mm) for solder dipped leadless chip carriers.

FIGURE 3 Physical dimensions of surface mount (UA versions). Similar to C-4.

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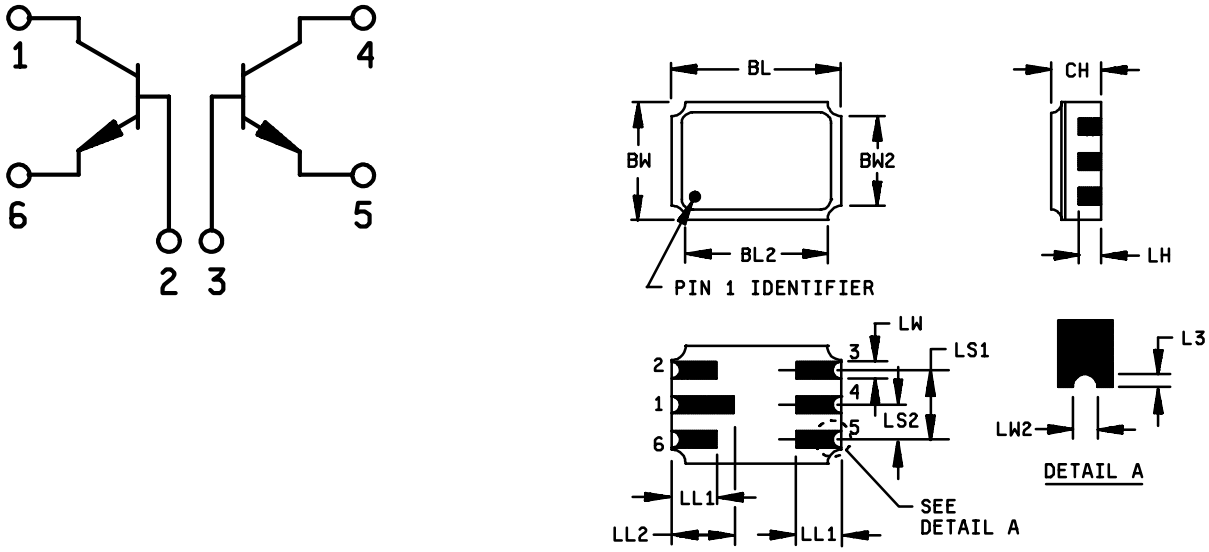


Inches	Millimeters	Inches	Millimeters
.016	0.41	.056	1.42
.017	0.43	.071	1.81
.022	0.56	.079	2.01
.024	0.61	.085	2.16
.035	0.89	.108	2.74
.038	0.96	.115	2.92
.039	0.99	.128	3.25
.046	1.17		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 4 Physical dimensions, surface mount (UB version) similar to C-3.

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Symbol	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
BL	.240	.250	6.10	6.35
BL_2		.250		6.35
BW	.165	.175	4.19	4.44
BW_2		.175		4.44
LW	.022	.028	0.56	0.71
LW_2	.006	.022	0.15	0.56
L_3	.003	.007	0.08	0.18
LH	.026	.034	0.66	0.86
LL_1	.060	.070	1.52	1.78
LL_2	.082	.098	2.08	2.49
LS_1	.095	.105	2.41	2.67
LS_2	.045	.055	1.14	1.39
CH	.066	.080	1.68	2.03

Transistor	Collector no. 1	Base no. 1	Base no. 2	Collector no. 2	Emitter no. 2	Emitter no. 1
Pin No.	1	2	3	4	5	6

Notes:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The coplanarity deviation of all terminal contact points, as defined by the device seating plane, shall not exceed .006 inch (0.15 mm) for solder dipped leadless chip carriers.

FIGURE 5 Physical dimensions, surface mount (U version) Similar to C-6

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3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in paragraphs table I herein.
. When a pacticular device is specified, the limit applies to all package types (e.g., 2N2369A, 2N2369AU, 2N2369AUA, and 2N2369AUB)

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
9	I_{CES} and h_{FE3}	Not applicable
11	I_{CES}; h_{FE3}; ΔI_{CES} = 100% of initial value or 25 nA dc, whichever is greater. Δh_{FE3} = ±15% of initial value.	I_{CES}, h_{FE3}
12	See 4.3.1 240 hours minimum	See 4.3.1 80 hours minimum
13	Subgroups 2 and 3 of table I herein; ΔI_{CES} = 100% of initial value or 25 nA dc, whichever is greater; Δh_{FE3} = ±15% of initial value.	Subgroup 2 of table I herein; ΔI_{CES} = 100% of initial value or 25 nA dc, whichever is greater; Δh_{FE3} = ±15% of initial value.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: T_A = Room ambient as defined in the general requirements of MIL-STD-750, paragraph 4.5;

2N2369A, 2N3227, 2N2369AU, 2N3227U, 2N2369AUA,
2N3227UA, 2N2369AUB, and 2N3227UB ---------- V_{CB} = 12 V dc, P_T = 360 mW.
2N4449, 2N4449U, 2N4449UA, and 2N4449UB---------- V_{CB} = 12 V dc, P_T = 300 mW.

NOTES:
1. No heat sink or forced air cooling on the devices shall be permitted during on cycle.
2. Power burn-in conditions apply for each section of the U version.

4.3.2 Thermal impedance ($Z_{\theta JX}$ measurements). The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, Method 3131.

a. I_M measurement current ---------- 5 mA.
b. I_H forward heating current ---------- 200 mA (min).
c. t_H heating time ---------- 25 - 30 ms.
d. t_{md} measurement delay time ---------- 60 µs max.
e. V_{CE} collector-emitter voltage ---------- 10 V dc minimum

The maximum limit for $Z_{\theta JX}$ under these test conditions is $Z_{\theta JX}$ (max) = 100°C/W.
The maximum limit for U package $Z_{\theta JX}$ (max) = 175°C/W per side
The maximum limit for UA package $Z_{\theta JX}$ (max) = 165°C/W per side
The maximum limit for UB package $Z_{\theta JX}$ (max) = 175°C/W per side

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed per MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied per 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500 and 4.4.2.1. Electrical measurements (end-points) and delta requirements shall be in accordance with group A, subgroup 2 and 4.5.3 herein. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 and shall be in accordance with group A, subgroup 2 herein. Delta measurements shall be in accordance with 4.5.3 herein.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 12 V dc; P_T = 360 mW for 2N2369A and 2N3227; V_{CB} = 12 V dc; P_T = 300 mW for 2N4449; T_A = +25°C ±3°C for all devices; $t_{on} = t_{off}$ = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on devices shall be permitted during on cycle.
B5	1027	V_{CB} = 12 V dc; T_A = +125°C ±25°C for 96 hours; P_T = 360 mW for 2N2369A and 2N3227, P_T = 300 mW for 2N4449 at T_A = +100°C or adjusted as required by the chosen T_A to give an average T_J = +275°C.

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV) Separate samples may be used for each step. In the event of a group B failure, the manufacturer may pull a new sample at double the sample size from either the failed assembly lot or from another assembly lot from the same wafer lot. If the new “assembly lot” option is exercised, the failed assembly lot shall be scrapped.

Step	Method	Condition
1	1039	Steady-state life: Test condition B, 340 hours, V_{CB} = 10 -30 V dc, T_J = 150°C min. No heat sink or forced-air cooling on the devices shall be permitted during on cycle. n = 45 devices, c = 0
2	1039	The steady state life test of step 1 shall be extended to 1,000 hrs for each die design. Samples shall be selected from a wafer lot every twelve months of wafer production. Group B step 2 shall not be required more than once for any single wafer lot. n = 45, c = 0.
3	1032	High-Temperature life (non-operating), T_A = +200°C. n = 22, c = 0

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL-PRF-19500.

b. Must be chosen from an inspection lot that has been submitted to and passed group A, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

4.4.3 Group C inspection, Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 (JANS).and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Electrical measurements (end points) and delta requirements shall be in accordance with group A, subgroup 2 and 4.5.3 herein.

4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E; not applicable for U, UA and UB devices.
C6	1026	1,000 hours at V_{CB} = 10 V dc; T_J = 150°C min. No heat sink or forced-air cooling on device shall be permitted during on cycle.

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4.4.3.2 Group C inspection, table VII (JAN, JANTX, and JANTXV) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E
C6		Not Applicable

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Input capacitance. This test shall be conducted in accordance with method 3240 of MIL-STD-750, except the output capacitor shall be omitted.

4.5.3 Delta Requirements. Delta requirements shall be as specified below:

Step	Inspection	MIL-STD-750		Symbol	Limit	Unit
		Method	Conditions			
1.	Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 10 mA dc; pulsed (see 4.5.1)	Δh_{FE3}	±25 percent change from initial value.	
2.	Collector - emitter and resistance	3071	I_C = 10 mA dc; I_B = 1.0 mA dc	$\Delta V_{CE(sat)}$	±50 mV change from previous measured value.	
3.	Collector - emitter cutoff current	3041	Bias condition C; V_{CE} = 20 V dc;	ΔI_{CES1}	100 percent of initial value or 25 nA dc, whichever is greater.	

1/ The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:

a. Subgroup 4, see 4.5.3 herein, step 2.

b. Subgroup 5, see 4.5.3 herein, steps 2, and 3.

2/ The electrical measurements for table VIb (JAN, JANTX and JANTXV) of MIL-PRF-19500 are as follows:

a. Subgroup 3 and 6, see 4.5.3 herein, step 1.

3/ The electrical measurements for table V of MIL-PRF-19500 are as follows:

a. Subgroup 6, see 4.5.3 herein, steps 1 and 3 (for JANS) and step 1 (for JANTX and JANTXV).

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TABLE I. Group A inspection

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical examination 3/	2071	n = 45 devices, c = 0				
Solderability 3/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/, 4/, 5/	1022	n = 15 devices, c = 0				
Temp Cycling 3/, 5/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
Heremetic Seal 5/	1071	n = 22 devices, c = 0				
Fine leak						
Gross leak						
Electrical measurements		Group A, subgroup 2				
Bond strength 3/	2037	Precondition T_A = +250°C at t = 24 hrs or T_A = 300°C at t = 2 hrs n = 11 wires, c = 0				
Subgroup 2						
Breakdown voltage, collector to base	3036	V_{CB} = 75 V dc	I_{CBO1}		10	µA dc
Breakdown voltage, emitter to base	3026		I_{EBO1}		10	µA dc
2N2369A, 2N4449		V_{EB} = 4.5 V dc				
2N3227		V_{EB} = 6.0 V dc				
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$			
2N2369A, 2N4449				15		V dc
2N3227				20		V dc
Collector to emitter cutoff Current	3041	Bias condition C; V_{CE} = 20 V dc	I_{CES}		0.4	µA dc
Collector to base cutoff Current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO2}		0.2	µA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 4 V dc	I_{EBO2}		0.25	µA dc
			h_{FE1}			
Forward-current transfer ratio	3076	V_{CE} = 0.35 V dc; I_C = 10 mA dc pulsed (see 4.5.1)				
2N2369A, 2N4449				40	120	
2N3227				70	250	
Forward-current transfer ratio	3076	V_{CE} = 0.4 V dc; I_C = 30 mA dc pulsed (see 4.5.1)	h_{FE2}			
2N2369A, 2N4449				30	120	
2N3227				40	250	
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 10 mA dc pulsed (see 4.5.1)	h_{FE3}			
2N2369A, 2N4449				40	120	
2N3227				100	300	

See footnotes at end of table.

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TABLE I. Group A inspection

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 100 mA dc; pulsed(see 4.5.1)	h_{FE4}			
2N2369A, 2N4449				20	120	
2N3227				30	150	
Forward-current transfer ratio	3076	V_{CE} = 10 V dc; I_C = 500 mA dc; pulsed see 4.5.1	h_{FE5}			
2N2369A, 2N4449				20		
2N3227				30		
Collector-emitter saturation voltage	3071	I_C = 10 mA dc; I_B = 1.0 mA dc pulsed (see 4.5.1)	$V_{CE(sat)1}$		0.20	V dc
Collector-emitter saturation voltage	3071	I_C = 3.0 mA dc; I_B = 3.0 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		0.25	V dc
Collector-emitter saturation voltage	3071	I_C = 100 mA dc; I_B = 10 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)3}$		0.45	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 10 mA dc; I_B = 1.0 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)1}$	0.70	0.85	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 30 mA dc; I_B = 3.0 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)2}$		0.90	
Base-emitter saturation voltage	3066	Test condition A; I_C = 100 mA dc; I_B = 10 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)3}$	0.80	1.20	V dc
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 20 V dc	I_{CBO2}		30	µA dc
High temperature operation		T_A = +125°C				
Collector to base cutoff current	3041	Bias condition A; V_{CE} = 10 V dc; V_{BE} = 0.25 V dc	I_{CEX2}		30	µA dc
Collector - emitter voltage saturated	3071	I_C = 10 mA dc I_B = 1.0 mA dc	$V_{CE(sat)4}$		0.3	V dc
Base - emitter saturated voltage	3066	Test condition A; I_C = 10 mA dc; I_B = 1.0 mA dc	$V_{BE(sat)4}$			
2N2369A, 2N4449				0.59		V dc
2N3227				0.50		V dc
Low temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 10 mA dc pulsed (see 4.5.1)	h_{FE5}			
2N2369A, 2N4449				20		
2N3227				40		
Base - emitter saturated voltage	3066	Test condition A; I_C = 10 mA dc; I_B = 1.0 mA dc	$V_{BE(sat)5}$		1.02	V dc

See footnotes at end of table.

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TABLE I. Group A inspection -Continued

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Small-signal short-circuit forward current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 10 mA dc; f = 100 MHz	$\lvert h_{fe} \rvert$	5.0	10	
Magnitude of small-signal short- circuit forward current transfer ratio	3306	V_{CE} = 20 V dc; I_C = 20 mA dc; f = 100 MHz	$\lvert h_{fe} \rvert$	2.5		
Open circuit Output capacitance	3236	V_{CB} = 5 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		4.0	pF
Input capacitance (output open- circuited)	3240	V_{EB} = 0.5 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz	C_{ibo}			
2N2369A, 2N4449					5.0	pF
2N3227					4.0	pF
Charge storage time		I_C = 10 mA dc; I_{B1} = 10 mA dc; I_{B2} = 10 mA dc; (see figure 1)	t_S			
2N2369A, 2N4449					13	ns
2N3227					18	ns
Turn-on time		I_C = 10 mA dc; I_{B1} = 3.0 mA dc; I_{B2} = 1.5 mA dc; (see figure 2)	t_{on}		12	ns
Turn-off time		I_C = 10 mA dc; I_{B1} = 3.0 mA dc; I_{B2} = 1.5 mA dc; (see figure 2)	t_{off}			
2N2369A, 2N4449					18	ns
2N3227					25	ns

NOTES:

1/ For sampling plan, see MIL-PRF-19500.

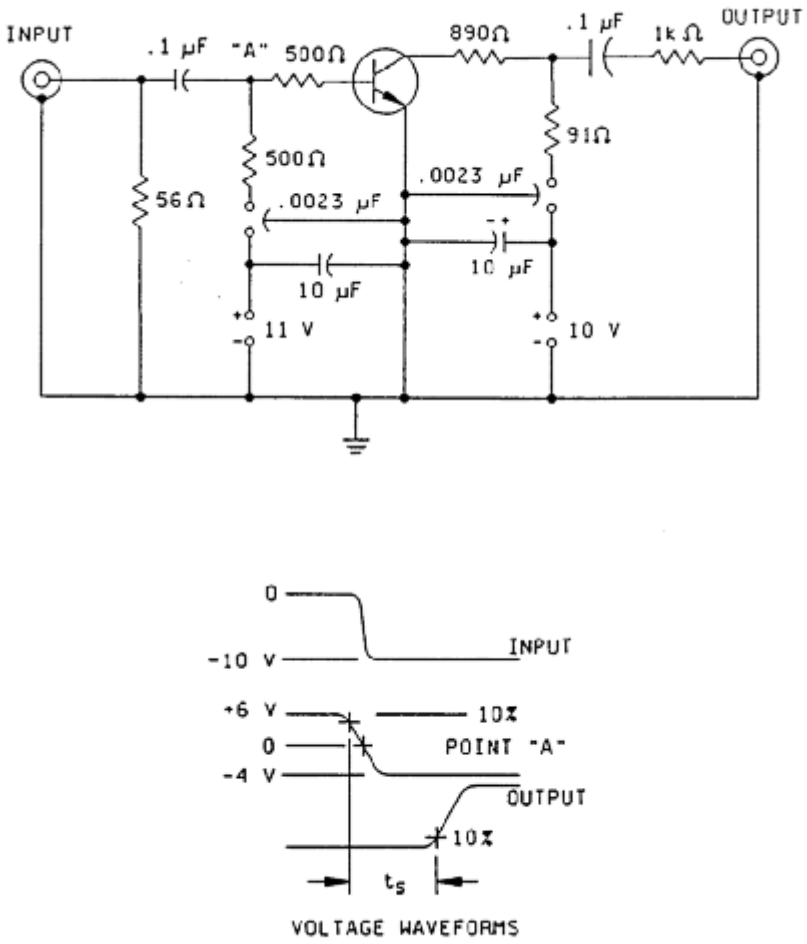
2/ For resubmission of failed subgroup A1, double the sample size of the failed test or sequence of tests. A failure in group A, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.

3/ Separate samples may be used.

4/ Not required for JANS devices.

5/ Not required for laser marked devices.

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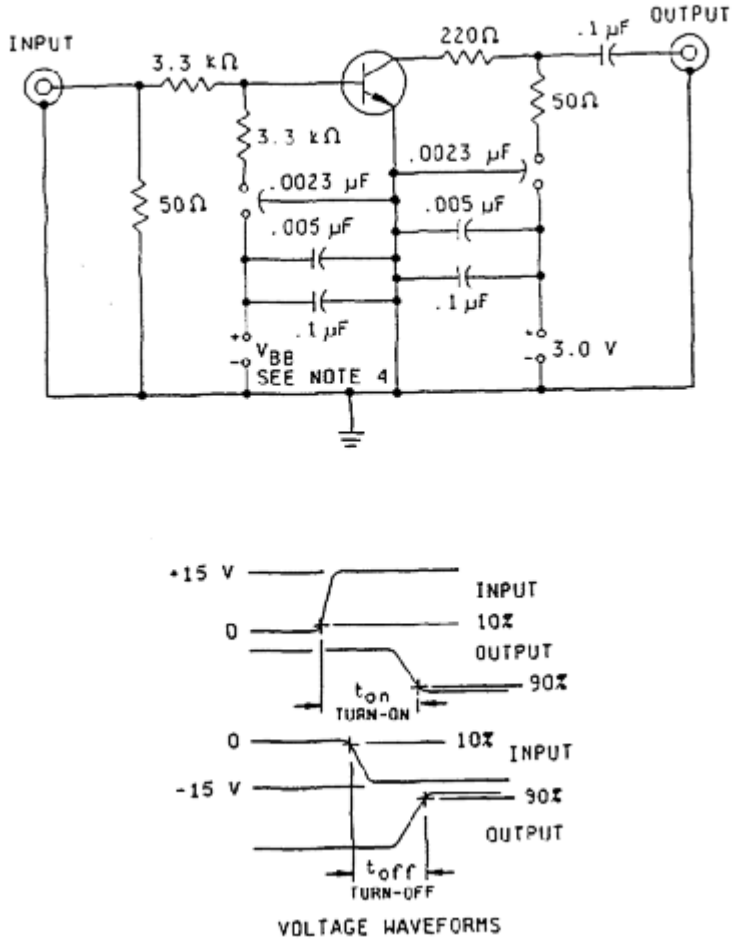


NOTES:

1. All capacitance in µF.
2. The input waveforms for each circuit are supplied by a pulsed generator with the following characteristics: Z_{OUT} = 500Ω, $t_r \leq 1$ ns, PW ≥ 300 ns, duty cycle ≤ 2 percent.
3. Output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 0.1$ ns, Z_{IN} = 50Ω.

FIGURE 4. Charge storage time test circuit.

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NOTES:

1. All capacitance in μF.
2. The input waveforms for each circuit are supplied by a pulse generator with the following characteristics: Z_{OUT} = 50Ω, $t_r \leq 1$ ns, PW ≥ 300 ns, duty cycle ≤ 2 percent.
3. Input and output waveforms are monitored on an oscilloscope with the following characteristics: $t_r \leq 0.1$ ns, Z_{IN} = 50Ω.
4. V_{BB} = -3.0 V for t_{on}, +12.0 V for t_{off}.

FIGURE 5. Turn-on and turn-off test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Lead finish may be specified (see 3.4.1).

b. Type designation and product assurance level.

c. Surface mount designation if applicable. (See figures 3, 4, and 5)

d. Packaging requirements (see 5.1).

6.3 Supersession information. Devices covered by this specification supersede the manufacturers' and users' Part or Identifying Number (PIN). The term Part or Identifying Number (PIN) is equivalent to the term part number which was previously used in this specification. This information in no way implies that manufacturers' PIN's are suitable as a substitute for the military PIN.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List QML-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA
DLA - CC

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 13, 19, 99

Preparing activity:
DLA - CC

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3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING TYPES 2N2369A, 2N2369AU, 2N2369AUA, 2N2369AUB, 2N3227, 2N3227U, 2N3227UA, 2N3227UB, 2N4449, 2N4449U, 2N4449UA, AND 2N4449UB, JAN, JANS, JANTX AND JANTXV

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

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8. PREPARING ACTIVITY

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c. ADDRESS (Include Zip Code) Defense Supply Center Columbus, ATTN: DSCC-VAC, 3990 East Broad Street, Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Vaginia 22060-6888 Telephone (703) 767-6888 DSN 427-6888

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